



FACULTY OF COMPUTER SCIENCE

BACHELORS DEGREE IN BUSINESS STUDIES AND COMPUTING SCIENCE

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COURSE: HCS213

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ASSIGNMENT 1

1)

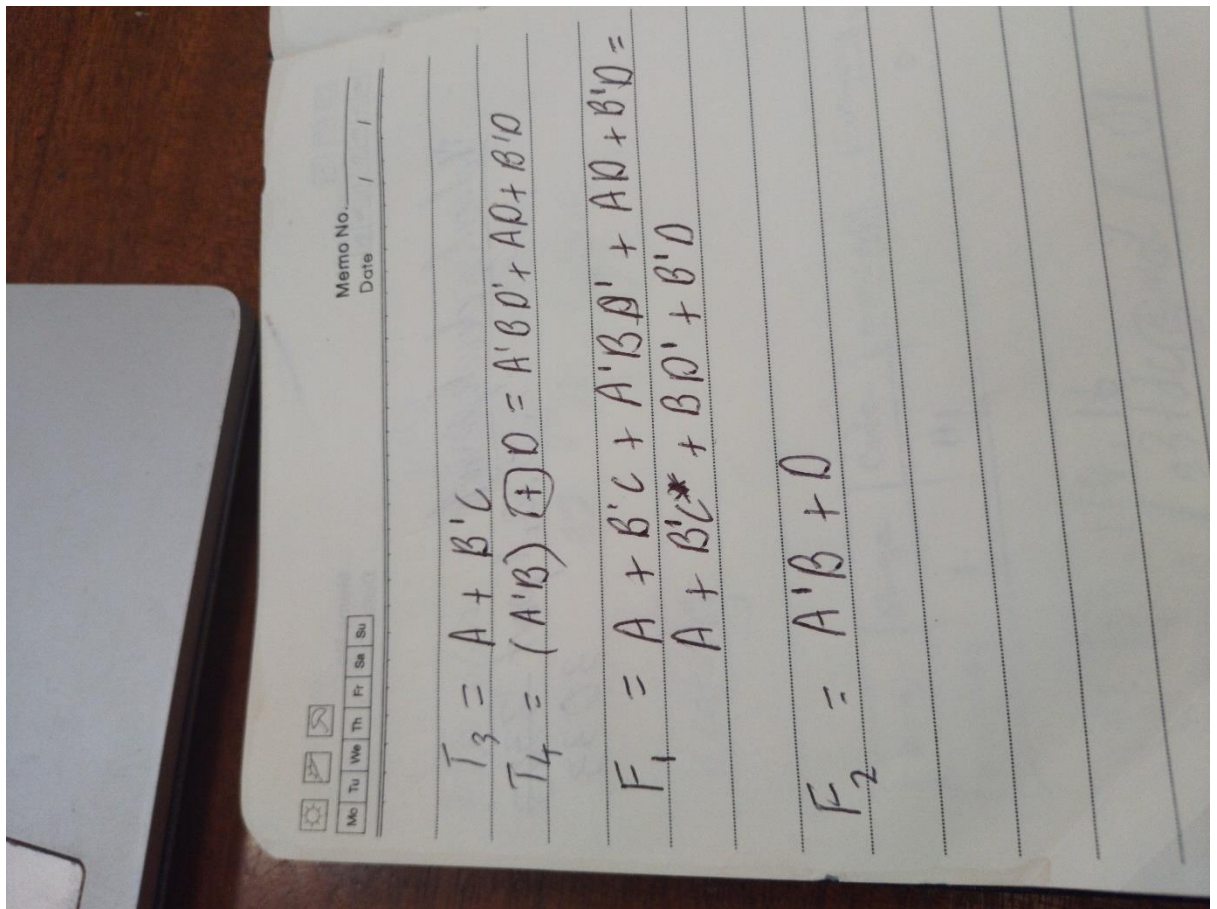
a)

i) $T1 = B'C$

ii) $T2 = A'B$

2)

b)

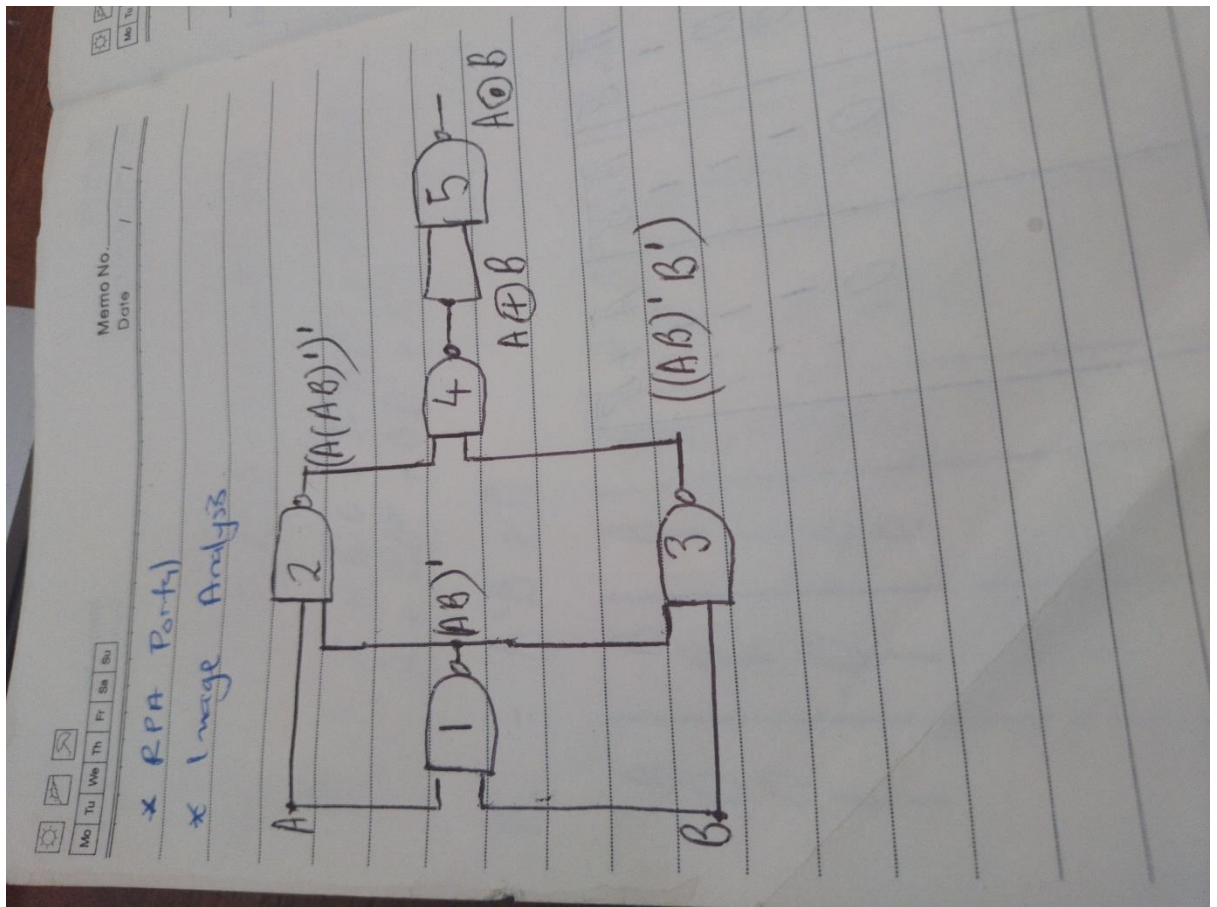


3) A) State and prove De Morgan's Theorem using truth tables.

A	B	$\bar{A}$	$\bar{B}$	$A+B$	$A \cdot B$	$\overline{A+B}$
0	0	1	1	0	0	1
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	1	1	0

$\bar{A} \cdot \bar{B}$	$\overline{A \cdot B}$	$\overline{A+B}$
1	1	1
0	1	1
0	1	1
0	0	0

b) Implement the Exclusive OR operation using NAND gates only.



- 4) A burglar alarm for a bank is designed so that it senses four input signal lines. Line A is from the secret control switch, line B is from a pressure sensor under a steel safe in a locked closet, line C is from a battery powered clock, and line D is connected to a switch on the locked closet door. The following conditions produce a logic 1 voltage on each line:

The statement “when the safe is moved and the control switch is closed” is represented by AB , “when the closet is opened after banking hours” is represented by $C'D$, “when the closet is opened with the control switch open” is represented by $A'D$ therefore $f(A,B,C,D) = AB' + C'D' + A'D$

5)

$$\begin{aligned}
 F &= \overline{A} \cdot \overline{A}D \cdot (\overline{A} + BC) & G &= (\overline{A} + BC)(\overline{A}D) \\
 &= (A + \overline{A}D) \cdot (\overline{A} + BC) & &= (\overline{A} + BC)(A + \overline{D}) \\
 &= (A + D)(\overline{A} + BC) & &= \overline{A}D + ABC + BC\overline{D} \\
 &= ABC + \overline{A}D + BC\overline{D}
 \end{aligned}$$

6)

- Explain how the following decimal-to-BCD encoder works.
- Determine which output (D or A) is the most significant bit of the BCD output.